Comparison of ML and DL models on Predicting Online Gaming Addiction in Bangladesh

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***Abstract*—Online gaming addiction remains a new problem in Bangladesh, but predictive studies in this field are normally Bangladesh-specific, and are often based on traditional machine learning (ML), baselines of which are not often reported using deep-learning (DL). The paper contains a systematic comparison of ML vs. DL in the task of predicting a binary addiction category (Class Y/N) of a survey dataset (N = 293; N = 170, Y = 123) in a Bangladesh-context setting. Two experimental conditions are considered: In Setting A all the features are available (behavioral-criteria variables are used), in Setting B lightweight variables only (Age, Gender, Gaming Frequency, and Opinion) are used. In order to guarantee that we are doing a fair evaluation, we employ leakage-safe preprocessing pipelines (imputation, one-hot encoding, scaling), a 80/20 stratified train-test split, and stratified 5-fold cross-validation on the training set. Accuracy, F1-score, ROC-AUC, and PR-AUC are used to report performance to take into consideration the issue of class imbalance. Setting A PyTorch MLP has the highest held-out performance (Accuracy = 0.949, F1 = 0.941, ROC-AUC = 0.989, PR-AUC = 0.988), and is closely followed by XGBoost (Accuracy = 0.949, F1 = 0.939). Setting B also shows significant performance decline (maximum F1 = 0.723), which demonstrates the significance of behavioral-criteria features. On the whole, the findings indicate that the DL models are competitive and should be taken into account in the future studies in the context of Bangladesh.**

***Index Terms*—Online gaming addiction, Bangladesh, tabular classification, machine learning, deep learning, MLP.**

I. Introduction

Online gaming is a common pastime among teenagers and young adults due to the rise in the use of smartphones, low-cost internet, and the development of competitive and social multiplayer games. Although gaming has the potential to provide entertainment and social interface, excessive and uncontrollable gaming can result in the clinically relevant impairment, such as disturbed sleeping and daily routines, poor academic/work performance, and worsening psychosocial health. Since most of these harms are cumulative, early diagnosis of the endangered people is a significant move towards awareness, screening and early intervention.

The studies on the addiction of online gaming in the Bangladesh setting are not extensive relative to the general studies carried out at the international level, and most of the available studies are based on the conventional machine learning (ML) pipelines. As an illustration, other researchers in Bangladesh have used numerous classical classifiers (e.g., Logistic Regression, Random Forest, Gradient Boosting, a version of XGBoost/AdaBoost) and found great performance of ensemble models like AdaBoost on survey based datasets [1]. Other publications focused on Bangladesh also highlight traditional ML workflows that use standard train-test splits and preprocessing [2]. Also, local surveys-based research indicates the pragmatic difficulties of gathering information and makes a direct reference to the necessity of utilizing more advanced ML and deep learning (DL) technologies to generate more compelling analytic results and standards [3]. In addition to Bangladesh, there are both behavioral and predictive perspectives of the literature. The studies involving large populations of MMORPG users investigate the prevalence of addiction based on validated criteria and investigate the predictors of behavioral (e.g., time spent playing) that could be associated with the risk of addiction [4]. Simultaneously, machine-learning-based detection and addiction-related applications have demonstrated effectiveness, such as the one that trains neural networks (e.g., CNN-based methods) on physiological data to classify addiction [5]. Nevertheless, even with this development, DL is not utilized extensively in gaming addiction prediction with Bangladesh in its scope, and no systematic, apples-to-apples comparisons is made between the use of machines learning and deep learning in these tasks in a consistent evaluation procedure.

Inspired by this gap, this paper provides a comparative benchmark of ML and DL models to predict online gaming

addiction with the help of a survey dataset that is relevant to Bangladesh (binary target: addicted vs. non-addicted). We consider two experimental conditions (Setting A) where all the available features are used (dimension of gaming addiction scale-style criteria and other survey attributes) and (Setting B) where predictors are limited to a small set of general attributes (e.g., Age, Gender, Frequency and Opinion, when included). Since the dataset has a mild-moderate imbalance in the classes, we report both threshold-dependent (e.g., F1-score) and ranking-based (ROC-AUC and PR-AUC) metrics, since the latter gives a more accurate representation of the performance when the (positive) class is underrepresented.

II. Literature Review

A theme identified in gaming addiction research is that the definition and measurement of addiction has direct effects on the prevalence estimates and the predictive modeling. The studies of MMORPG addiction demonstrate that the application of various classification criteria can result in highly diverse rates of addiction even on the same data, and that the symptom-based criteria should be used with care and transparency by model developers [4]. This is in agreement with the Setting A that uses information-laden behavioral cues as opposed to surface proxies. Survey-based work also reports that gaming addiction has been acknowledged as a mental-health problem (e.g. ICD-11 "Gaming Disorder"), but the field frequently does not have standard datasets, so researchers tend to develop their own [3]. It is thus possible to extract more clinically significant evidence of questionnaire data using structured behavioral indicators (Setting A).

The common type of modeling usually depends on features that are lightweight and easily gathered, in terms of age, gender, play time, and gaming history. The analysis of MMORPG based on logistic regression indicates that the employment, the number of years playing, and the hours spent playing predict the higher-risk addiction level, which is also aligned with the minimal level of demographic/usage variables of the Setting B [4]. A study of mobile game addiction in university students conducted in Bangladesh also implicates demographic and contextual (e.g., gender) and psychosocial (e.g., loneliness) factors in predicting the level of addiction, where play duration and time-of-day patterns are also important variables, making them predictable in some ways because they are easy to collect and analyze [6]. This is further developed by other studies that add health and lifestyle variables to predict digital game addiction, including age, gender, BMI, and physical activity, which is a more inclusive proxy-feature design that is closer to Setting B than symptom-based modeling [7].

A second stream of research includes psychological and engagement constructs to more effectively explain problematic use. A consumer-engagement study investigates the mediation of such dimensions as immersion and pleasure between addiction development, which is conceptually related to the phenomenon of Setting A since it clarifies proxies of deeper behavioral inclinations [8]. New methods also move predominantly away tabular surveys, towards unstructured behavioral traces, like gamer-text. The GAMEADDICTED system is based on tagged user-reviews on addiction cues and it shows high accuracy with boosted classifiers, such as XGBoost, which proves that addiction cues can be detected with the use of language and that advanced ML systems hold promise [9].

In Bangladesh, there are numerous studies that are based on ML, most with only one train/test split and report high accuracy, but without stressing the importance of dataset size, labeling, and evaluation processes. These results underline the necessity of standardized testing and a careful generalization, which can encourage a systematic comparison of feature settings (Setting A vs. B) and model families (ML vs. DL) in the context of a standardized protocol [2]. The presence of practical data problems and the necessity to use structured datasets should also justify feature-split methodology that was chosen in the present study and be appreciated in the future studies in Bangladesh [3].

III. Methodology

We base our work on previous research that mostly focuses on classical ML models and usually uses a small set of features or a single split of datasets, but we develop a common experimental design to make the comparison between classic ML and deep learning to predict gaming addiction in the context of Bangladesh. We compare two feature settings - Setting A (full behavioral + demographic/usage attributes) and Setting B (lighweight demographic/usage attributes only) - under leakage-safe preprocessing, stratified cross-validation with training data, and the final evaluation with a held-out stratified split. This approachology will allow a controlled evaluation of the worth of behavioral-criteria features as well as competitiveness of DL baselines to developed ML methods.

A. Data Collection

Our data source is a publicly available dataset, which was able to be retrieved on Kaggle and includes survey responses gathered in the Bangladesh setting to analyze online gaming addiction. The data includes 293 cases and 12 variables, of which 11 are predictors and the target variable is Class having two categories (N = non-addicted, Y = addicted). Class is represented as a binary label (N-0, Y-1) in our experiments. The distribution of classes is also fairly unbalanced (N=170, Y=123) (Figure: 1). It is presented as a binary classification problem with the attributes of the survey, it is required to determine whether a participant is a member of an addicted class.

B. Experimental Feature Settings

To study the importance of behavioral-criteria variables versus lightweight demographics/usage variables, we evaluate two feature configurations:

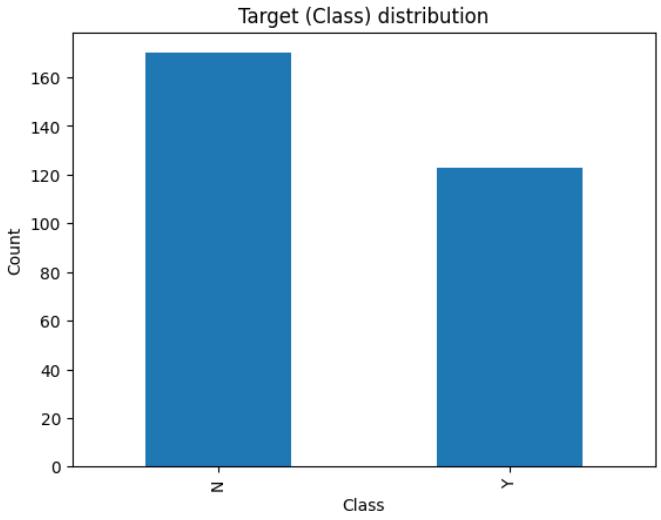


Fig. 1. Target class distribution.

- Setting A (Full Features): Uses all available predictors (11 features). This includes demographic/usage variables (e.g., Age, Gender, Frequency, Opinion) as well as behavioral criteria-style attributes (e.g., Salience, Tolerance, Mood Modification, Withdrawal, Relapse, Conflict, Problems).
- Setting B (Lightweight Features): Uses only Age, Gender, Frequency, Opinion, representing a low-cost screening scenario where detailed behavioral criteria may not be available.

This A/B design directly supports the paper's objective which is to quantify how much predictive power comes from richer behavioral criteria compared to general demographic/usage signals.

C. Data Preprocessing

The entire preprocessing is realized with a leakage-safe pipeline such that all transformations are only fitted to the training data and applied to validation/test data. Imputation of the numbers features using the median and non-numeric features using the most common category is the way missing values are done. One-hot encoding is used to convert categorical variables to numeric and z-score normalization (standard scaling) is used to standardize numeric variables. These changes are implemented using a ColumnTransformer that operates numeric and categorical columns one at a time and inputs the results into a single model as a part of a single Pipeline. The dataset is preprocessed and the evaluation is performed with a two stage protocol; an 80/20 stratified hold-out split is done to obtain a final test set to report on, and the training set is then evaluated using stratified 5-fold cross-validation, to maintain class proportions in each fold. In the case of deep learning models, a validation data is again sampled out of the training set to allow it to terminate early whereas final results are still reported on the same held-out test set so that the results can still be compared to the machine learning baselines.

D. Implemented Models

In an attempt to achieve a thorough and impartial analysis on this tabular binary classification problem, we compare a heterogeneous collection of both classical machine-learning (ML) and deep-learning (DL) models, including linear and probabilistic models, kernel-based models, instance-based models, as well as tree/ensemble models. This design enables us to examine model expressiveness as well as model robustness under a common preprocessing protocol and validation protocol.

i. Classical Machine-Learning Baselines

We incorporate low-variance linear and probabilistic Baselines such as Logistic Regression (LogReg) and Gaussian Naive Bayes (GNB) which are good points of reference when working with tabular data. In order to find non-linear decision boundaries, we consider Support Vector Machines using an RBF kernel (SVM-RBF) and k-Nearest Neighbors (KNN), which are a representation of a kernel based and distance based learning, respectively. As a non-linear baseline, an interpretable Decision Tree (DT) is provided. To extend our benchmarking of commonly used ensemble techniques on structured data, we compare Random Forest (RF) and Extra Trees (ET) as bagging-based techniques, and Gradient Boosting (GB), AdaBoost (Ada) and XGBoost (XGB) as boosting-based techniques. These models are established with respect to feature interaction and heterogeneous tabular input.

All the ML models are implemented with scikit-learn with Logistic Regression being set to an L2-regularized linear solver, SVM that uses an RBF kernel with probability outputs on and KNN which uses distance-weighted Euclidean neighborhoods. The moderately depth-constrained tree-based models are employed to minimize overfitting, and the multiple estimators of ensemble methods are employed to guarantee stability. To overcome the issue of class imbalance, all the supported models are trained with class-balanced weights, whereas XGBoost estimates a scale_pos_weight based on the proportion of the negative and positive samples in the training set. All ML models are trained on a leakage-safe pipeline, with a ColumnTransformer-based preprocessor being fed into the chosen estimator. Model assessment is performed based on stratified 5-fold cross-validation of the training split, and then on a stratified test set held out.

ii. Deep-Learning Baselines

Besides a classical ML model, we also apply three PyTorch-based feed-forward neural networks to determine whether basic neural architecture can make it competitive on this comparatively small tabular dataset. TorchMLP is the first model, it is a small multi-layered perceptron consisting

of two hidden layers, ReLU activations, and dropout as regularization measures. TorchMLP_BN, the second model, adds more capacity to the model and uses Batch Normalization prior to activation and dropout in every hidden layer to enhance the stability of optimization. TorchResMLP, the third model, adds residual connections with stacked residual blocks to flow gradients and improve representational capacity of the model to capture complex interactions of features.

Every neural model is trained on binary cross-entropy with logits (BCEWithLogitsLoss). In order to solve the class imbalance, a positive-class weight (pos_weight) is calculated based on training-set label distributions and added into the loss. Adam optimizer, weight decay, mini-batch training and constant learning rate are used to perform optimization. The training is done until up to 200 epochs and early stopping on validation loss to avoid overfitting. The logits are converted into probabilities during inference with a sigmoid function and classification measures are computed with a fixed threshold of 0.5.

E. Evaluation Metrics

In order to evaluate performance in moderate classes imbalance, we declare the following measures. Accuracy is used to measure the general proportion of sample that is correctly classified but not much informative when the classes are uneven. Precision measures the reliability to make positive predictions (i.e. the fraction of predicted cases of addicted individuals which are actually addicted), and Recall the sensitivity to detect addicts (i.e. the fraction of addicts correctly predicted). The harmonic mean of precision and recall is called F1-score and gives a single summary that punishes either extreme of the trade-off between the two. ROC-AUC compares the ranking capability of the model at each threshold by summing up the trade-off between the true rate of positive and false rate of positive. PR-AUC totals the precision-recall curve at each of the thresholds and is especially informative when the prevalence of the positive class is low, because the performance of the positive class is in focus and the presence of many negatives does not affect the result.

F. Additional Unsupervised Analysis

We also use K-Means (k=2) and silhouette to check the explorative validity of this difference through using two groups of samples lying naturally apart based on the criterion of addiction vs. non-addiction. It is an analysis that is applied to gain a deeper understanding rather than a forecast.

RESULT & DISCUSSION

This section presents a performance that was conducted based on the evaluation procedure applied in our code: Stratified 5-fold CV on the training split and final reporting on a held-out stratified test set (n = 59). Our report is Accuracy, Precision, Recall, F1, and threshold-free ROC-AUC and PR-AUC.

TABLE I
SETTING A (ALL FEATURES): HELD-OUT TEST RESULTS ORDERED BY F1. (* DENOTES CLASS-WEIGHTED MODELS.)

Model	Acc	Prec	Rec	F1	ROC	PR
MLP	0.949	0.923	0.960	**0.941**	0.989	0.988
XGB	0.949	0.958	0.920	0.939	0.986	0.983
MLP-BN	0.949	0.958	0.920	0.939	0.989	0.988
LogReg*	0.932	0.889	0.960	0.923	0.992	0.990
Ada	0.932	0.920	0.920	0.920	0.989	0.986
SVM-RBF*	0.932	0.957	0.880	0.917	0.985	0.984
GB	0.915	0.917	0.880	0.898	0.982	0.980
ET*	0.915	0.955	0.840	0.894	0.966	0.961
RF*	0.915	1.000	0.800	0.889	0.968	0.964
ResMLP	0.864	0.870	0.800	0.833	0.976	0.968
KNN(7)	0.831	0.857	0.720	0.783	0.935	0.933
DT*	0.797	0.882	0.600	0.714	0.853	0.832
GNB	0.780	0.929	0.520	0.667	0.849	0.865

A. Results for Setting A

Most models with the complete set of features (demographic/usage and behavioral-criteria variables) have high discrimination (Table: I). The optimum models achieve test accuracy of approximately 0.95 and test F1 of approximately 0.94, whereas ROC-AUC and PR-AUC are near 0.99, which means that label of addiction is highly predictable when behavioral-criteria features are added.

Confusion matrix insight (best model): The best-performing model by test F1, TorchMLP, produced the following confusion matrix on the 59-sample test set:

$$\text{TN} = 32,\ \text{FP} = 2,\ \text{FN} = 1,\ \text{TP} = 24$$

This indicates both low false-alarm rate and low miss rate when full behavioral information is available.

Cross-validation vs test behavior: TorchResMLP had the best CV mean F1 (0.945) and considerably lower test F1 (0.833) in Setting A. This information indicates that larger variance / overfitting probability of the more expressive residual model in a smaller dataset will be possible, even with early stopping. However, the simpler MLPs (TorchMLP and TorchMLP_BN) are more consistently generalizing.

B. Results for Setting B

In the case of using demographic/usage attributes only (Setting B), the performance of all models makes a significant drop (Table: II). The optimum test F1 equals 0.723 and the ROC-AUC values are approximately 0.76-0.79, which means that the predictive power of lightweight variables is moderate.

TABLE II
SETTING B (AGE, GENDER, FREQUENCY, OPINION): HELD-OUT TEST RESULTS ORDERED BY F1. (* DENOTES CLASS-WEIGHTED MODELS.)

Model	Acc	Prec	Rec	F1	ROC	PR
Ada	0.780	0.773	0.680	**0.723**	0.794	0.790
LogReg*	0.746	0.679	0.760	0.717	0.779	0.792
GNB	0.729	0.645	0.800	0.714	0.775	0.802
ResMLP	0.763	0.739	0.680	0.708	0.761	0.781
MLP	0.746	0.708	0.680	0.694	0.762	0.792
MLP-BN	0.746	0.708	0.680	0.694	0.749	0.790
SVM-RBF*	0.712	0.654	0.680	0.667	0.731	0.690
DT*	0.695	0.630	0.680	0.654	0.717	0.623
KNN(7)	0.627	0.545	0.720	0.621	0.697	0.683
ET*	0.610	0.529	0.720	0.610	0.696	0.671
RF*	0.610	0.531	0.680	0.596	0.710	0.723
GB	0.627	0.556	0.600	0.577	0.725	0.657
XGB	0.593	0.516	0.640	0.571	0.746	0.741

Confusion matrix insight (best model): The best Setting B model by test F1, AdaBoost, yields:

$$\text{TN} = 29,\ \text{FP} = 5,\ \text{FN} = 8,\ \text{TP} = 17$$

Notably, false negatives (8) are higher than false positives (5), meaning the lightweight model is more likely to miss addicted individuals—important if the application is screening.

Metric trade-offs: Whereas AdaBoost offers the highest F1, GaussianNB offers the highest PR-AUC (0.802) and the best recall (0.80) of the best models, the precision of the top models is lower. This demonstrates that using limited features, model choice might alter the tradeoff between the desired degree of positives to capture and the desired degree of false alarms.

C. Discussion

The most consistent result of the study is the A/B feature-setting comparison based on which, with all features (Setting A), the best model has F1 = 0.941 and ROC-AUC near 0.99, whereas with lightweight variables (Setting B) only, the performance decreases to F1 = 0.723 and ROC-AUC close to 0.79, thus suggesting that the behavioral-criteria variables included in the first feature-setting are the ones that contain the largest share of the discriminative signal, while the demograph Deep learning is obviously competitive in Setting A, where even a simple TorchMLP is best at test F1 (0.941) and test accuracy (0.949), and deep baselines such as XGBoost and class-weighted Logistic Regression are highly effective, despite slightly worse F1. Setting B There is no deep learning dominance; AdaBoost produces the highest test F1, and the DL models are similar but a little lower, indicating that in cases of small feature space and the large share of low-level features, classical methods may be adequate. We can also note that certain advanced-capacity models (especially TorchResMLP in Setting A) exhibit the difference between cross-validation and held-out test performance which is caused by the increase of variance/overfitting risk on smaller datasets and it is important to report both CV and test results and use larger samples or repeated splits in the future. On deployment, the Logistic Regression model has a good performance, interpretability, and low-computation cost, as compared to boosting/ensemble (AdaBoost/XGBoost) which has good non-linear modeling but requires further explainability tools to be transparent; the MLP-based DL models, though necessitating class-weighted loss and early stopping, demonstrates the ability to be as competitive as market-leading ML baselines when richer behavioral features are provided, hence the rationale as to why DL should be considered in Bangladesh-context studies. Lastly, an unsupervised KMeans (k=2) exploratory analysis yielded a small silhouette score (0.0973, Davies-Bouldin 2.9365), and did not cluster visibly with the labels of addictions, which again demonstrates the importance of labeled learning when using behavioral-criteria features.

CONCLUSION & FUTURE WORK

A benchmark on the prediction of online gaming addiction based on both traditional machine learning (ML) and deep learning (DL) methods on a survey dataset publicly available (N = 293; Class: N = 170, Y = 123) in the context of Bangladesh was provided in the paper. In order to measure the value of the various types of features, we tested two different settings: Setting A (when the features being used are behavioral-criteria features) and Setting B (when the features used are lightweight features (Age, Gender, Frequency, Opinion)). We identified behavioral-criteria features to be vital to accurate prediction using leakage-safe cross-validation pipelines, stratified 5 fold cross-validation on the training set, and final evaluation on a held-out stratified test split. Setting A In Setting A, a PyTorch MLP had the best held-out performance (Acc = 0.949, F1 = 0.941, ROC-AUC = 0.989, PR-AUC = 0.988) and was comparable to state-of-the-art ML baselines like XGBoost. Setting B Performance was much lower (maximum F1 = 0.723), demonstrating that demographics and general usage variables do not achieve high-confidence addiction prediction in this data set. All in all, the findings attest to the key hypothesis of the research, i.e., deep learning baselines can be competitive in the context of Bangladesh and must be taken into account together with more traditional ML strategies, in particular when more behavioral information is accessible.

The suggested approach is to be tested in future on bigger and more diverse datasets representing Bangladesh context and external validation (or repeated stratified splits) should be employed to get more consistent generalization estimates. It would also help to research probability calibration and

threshold choice to deployment oriented on screening, and include interpretability analysis (e.g., feature attribution) to better understand which behavioral factors are driving the predictions. In addition to MLP baselines, future research can compare modern tabular deep learning models to strong tree ensembles and investigate more modalities like patterns of long-term usage or Bangali text cues to the extent that is ethically sound. Lastly, these models must be regarded as screening models and not clinical diagnostic models and must be implemented with adequate monitoring and protective measures.

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